

Open Addressing — Exercises 11.4

Exercise 11.4-1

Insert the keys

10, 22, 31, 4, 15, 28, 17, 88, 59

into a hash table of size $m = 11$ using:

- Linear probing
- Quadratic probing with $c_1 = 1$ and $c_2 = 3$
- Double hashing with

$$h_1(k) = k \bmod 11, \quad h_2(k) = 1 + (k \bmod 10)$$

Show the probe sequence for every key and give the final hash table for each technique.

Solution

For all three methods, the auxiliary hash function is

$$h'(k) = k \bmod 11.$$

The keys are inserted in the given order.

1. Linear Probing

The probe function is

$$h(k, i) = (h'(k) + i) \bmod 11, \quad i = 0, 1, 2, \dots$$

If the first position is occupied, we move to the next slot, wrapping around modulo 11.

Key	$h'(k)$	Probe sequence	Final slot
10	10	10	10
22	0	0	0
31	9	9	9
4	4	4	4
15	4	4 → 5	5
28	6	6	6
17	6	6 → 7	7
88	0	0 → 1	1
59	4	4 → 5 → 6 → 7 → 8	8

For example, key 59 starts at

$$59 \bmod 11 = 4.$$

Slots 4, 5, 6, 7 are already occupied, so it is placed in slot 8.

Final table — Linear Probing

Slot	0	1	2	3	4	5	6	7	8	9	10
Key	22	88	–	–	4	15	28	17	59	31	10

2. Quadratic Probing

With $c_1 = 1$ and $c_2 = 3$,

$$h(k, i) = (h'(k) + i + 3i^2) \bmod 11.$$

Key	$h'(k)$	Probe sequence	Final slot
10	10	10	10
22	0	0	0
31	9	9	9
4	4	4	4
15	4	$4 \rightarrow 8$	8
28	6	6	6
17	6	$6 \rightarrow 10 \rightarrow 9 \rightarrow 3$	3
88	0	$0 \rightarrow 4 \rightarrow 3 \rightarrow 8 \rightarrow 8 \rightarrow 3 \rightarrow 4 \rightarrow 0 \rightarrow 2$	2
59	4	$4 \rightarrow 8 \rightarrow 7$	7

A few nontrivial insertions are:

$$\begin{aligned} h(17, 0) &= (6 + 0 + 0) \bmod 11 = 6, \\ h(17, 1) &= (6 + 1 + 3) \bmod 11 = 10, \\ h(17, 2) &= (6 + 2 + 12) \bmod 11 = 9, \\ h(17, 3) &= (6 + 3 + 27) \bmod 11 = 3, \end{aligned}$$

so 17 is placed in slot 3.

For 59,

$$\begin{aligned} h(59, 0) &= 4, \\ h(59, 1) &= (4 + 1 + 3) \bmod 11 = 8, \\ h(59, 2) &= (4 + 2 + 12) \bmod 11 = 7, \end{aligned}$$

and slot 7 is empty, so 59 is inserted there.

Final table — Quadratic Probing

Slot	0	1	2	3	4	5	6	7	8	9	10
Key	22	–	88	17	4	–	28	59	15	31	10

Important: with the stated constants $c_1 = 1$, $c_2 = 3$, and $m = 11$, key 59 *does not fail*; it is inserted into slot 7.

3. Double Hashing

The probe function is

$$h(k, i) = (h_1(k) + i h_2(k)) \bmod 11,$$

where

$$h_1(k) = k \bmod 11, \quad h_2(k) = 1 + (k \bmod 10).$$

Because 11 is prime and every $h_2(k)$ lies in $\{1, 2, \dots, 10\}$, every step size is relatively prime to 11. Thus a probe sequence can visit all table positions if necessary.

Key	$h_1(k)$	$h_2(k)$	Probe sequence	Final slot
10	10	1	10	10
22	0	3	0	0
31	9	2	9	9
4	4	5	4	4
15	4	6	$4 \rightarrow 10 \rightarrow 5$	5
28	6	9	6	6
17	6	8	$6 \rightarrow 3$	3
88	0	9	$0 \rightarrow 9 \rightarrow 7$	7
59	4	10	$4 \rightarrow 3 \rightarrow 2$	2

For example, for key 15,

$$h_1(15) = 4, \quad h_2(15) = 1 + (15 \bmod 10) = 6.$$

Hence

$$\begin{aligned} h(15, 0) &= (4 + 0 \cdot 6) \bmod 11 = 4, \\ h(15, 1) &= (4 + 1 \cdot 6) \bmod 11 = 10, \\ h(15, 2) &= (4 + 2 \cdot 6) \bmod 11 = 5. \end{aligned}$$

Slots 4 and 10 are occupied, so 15 is placed in slot 5.

For key 59,

$$h_1(59) = 4, \quad h_2(59) = 1 + (59 \bmod 10) = 10,$$

so the probes are

$$4 \rightarrow 3 \rightarrow 2,$$

and 59 is placed in slot 2.

Final table — Double Hashing

Slot	0	1	2	3	4	5	6	7	8	9	10
Key	22	–	59	17	4	15	28	88	–	31	10

Exercise 11.4-2

Write pseudocode for HASH-DELETE and modify HASH-INSERT to handle DELETED.

Solution

Algorithm 1 HASH-DELETE(T, k)

```
 $i \leftarrow 0$ 
repeat
   $j \leftarrow h(k, i)$ 
  if  $T[j] = k$  then
     $T[j] \leftarrow \text{DELETED}$ 
    return  $j$ 
  end if
   $i \leftarrow i + 1$ 
until  $T[j] = \text{NIL}$  or  $i = m$ 
error
```

Algorithm 2 HASH-INSERT(T, k)

```
 $i \leftarrow 0$ 
repeat
   $j \leftarrow h(k, i)$ 
  if  $T[j] = \text{NIL}$  or  $T[j] = \text{DELETED}$  then
     $T[j] \leftarrow k$ 
    return  $j$ 
  end if
   $i \leftarrow i + 1$ 
until  $i = m$ 
error
```

Exercise 11.4-3

Give expected probe bounds when $\alpha = 3/4$ and $\alpha = 7/8$.

Solution

Using Theorem 11.6:

$$E[\text{unsuccessful}] = \frac{1}{1 - \alpha}$$

Using Theorem 11.8:

$$E[\text{successful}] = \frac{1}{\alpha} \ln \frac{1}{1 - \alpha}$$

$$\alpha = 3/4$$

Unsuccessful:

$$\frac{1}{1 - 3/4} = 4$$

Successful:

$$\frac{1}{3/4} \ln 4 \approx 1.848$$

$$\alpha = 7/8$$

Unsuccessful:

$$\frac{1}{1 - 7/8} = 8$$

Successful:

$$\frac{1}{7/8} \ln 8 \approx 2.377$$

Exercise 11.4-4

★ Show that if $d = \gcd(m, h_2(k)) \geq 1$, then probe sequence examines only m/d slots.

Solution

Let:

$$h(k, i) = (h_1(k) + ih_2(k)) \bmod m$$

Let $d = \gcd(m, h_2(k))$.

Then $h_2(k)$ generates a cyclic subgroup modulo m of size:

$$\frac{m}{d}.$$

Thus the probe sequence repeats after m/d steps.

If $d = 1$, the sequence has full period m .

If $d > 1$, only m/d slots are examined.

Exercise 11.4-5

★ Find α such that unsuccessful search equals twice successful search.

Solution

Solve:

$$\frac{1}{1-\alpha} = 2 \cdot \frac{1}{\alpha} \ln \frac{1}{1-\alpha}$$

Numerical solution:

$$\boxed{\alpha \approx 0.71533}$$